

Cross-city Time Series Forecasting with Retrieval-Augmented Large Language Models

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Abstract

The World Wide Web increasingly relies on intelligent services that require accurate time series forecasting, from urban mobility platforms to adaptive web-based decision systems. In practice, building effective forecasting models typically requires abundant high-quality data, which may not always be available in all cities due to sensing limitations or data sparsity. To address this challenge, transfer learning methods aim to transfer knowledge from data-rich source cities to data-scarce target cities. However, source and target data distributions are often not identical: while some patterns from source cities may be beneficial, others can be irrelevant or even misleading. Existing transfer learning methods generally train the target model using all available source data without explicitly distinguishing between useful and non-useful knowledge, which may hinder performance. In this work, we propose **xRAG4TS**, a novel framework that integrates Retrieval-Augmented Generation (RAG) with Large Language Models (LLMs) for cross-city time series forecasting. xRAG4TS introduces a *Cross-City Selective Retriever Module* that filters semantically relevant historical patterns and documents from data-rich source cities, and incorporates them as structured prompts in an *LLM Inference Module* to guide forecasting in data-scarce target cities. By enabling selective, interpretable, and context-aware knowledge transfer, our method enhances robustness and scalability in web-oriented spatio-temporal applications. Extensive experiments on real-world cross-city datasets demonstrate that xRAG4TS significantly outperforms state-of-the-art baselines, highlighting its potential for powering adaptive and trustworthy web services under severe data scarcity.

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CCS Concepts

• **Information systems** → **Spatial-temporal systems**; • **Computing methodologies** → **Machine learning**.

Keywords

Urban Computing, Time Series Forecasting, Retrieval-Augmented Generation, Large Language Models, Knowledge Transfer

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1 Introduction

Time series forecasting underpins a wide range of intelligent web services, powering applications such as traffic-aware navigation, parking availability platforms, personalized mobility recommendations, and energy-efficient urban dashboards [6, 8–10, 14, 19, 47]. As the scale and complexity of web-based urban systems grow, accurate forecasting becomes a cornerstone for adaptive and data-driven decision making. Recent advances in deep learning, particularly sequence modeling architectures [24, 44, 53] and spatio-temporal graph neural networks (STGNNs) [1, 20, 34, 45, 51], have demonstrated strong capability in modeling complex temporal dependencies and spatial interactions. However, the deployment of such models in real-world web environments faces a critical bottleneck: the availability of large-volume and fine-grained data. In many practical scenarios, data scarcity arises from limited sensing infrastructure, privacy restrictions, or the early stages of system deployment. This challenge manifests in both spatial dimensions (e.g., insufficient monitored locations) and temporal dimensions (e.g., lack of long-term historical data), severely restricting the generalizability and effectiveness of standard forecasting models.

A promising solution to this challenge is cross-city transfer learning, which aims to leverage data-rich source cities to improve forecasting performance in data-scarce target cities. The core idea is to transfer useful representations, knowledge, or model parameters learned in the source domain to aid the target domain, ideally without requiring extensive labeled data in the latter. Prior works have explored this direction through pretraining-finetuning pipelines, shared embedding spaces, and transferable graph representations [16, 17, 27, 40].

However, these methods still face two major limitations. First, they often overlook the node-level spatio-temporal heterogeneity between source and target cities, which can lead to sub-optimal graph learning between cities and degraded knowledge transfer. In practice, cities exhibit substantial heterogeneity in spatial configurations, traffic dynamics, temporal usage patterns, and operational characteristics. As a result, many time series in the source city may be unrelated or even harmful to the target forecasting task. Their inter- and intra-correlations with other entities within and across cities can vary significantly, leading to divergent graph structures. Existing transfer learning methods [17, 27] indiscriminately migrate global forecasting models or full graph representations from the source to the target domain without explicit relevance filtering. This often introduces spatio-temporal noise, resulting in negative transfer effects, where irrelevant or misleading source patterns degrade forecasting accuracy in the target city [13].

Second critical limitation of existing transfer learning frameworks lies in their reliance on black-box neural adaptation, which limits interpretability and undermines user trust in real-world applications [5, 30, 31]. These methods typically adjust model parameters or hidden representations without providing insight into how transferred knowledge informs the prediction. This lack of transparency is particularly problematic in high-stakes urban forecasting scenarios, where decision-makers require justifiable and causally grounded outputs. Moreover, these models rarely incorporate commonsense reasoning or semantic context.

To address these limitations, we propose xRAG4TS, a novel Retrieval-Augmented Generation framework tailored for cross-city time series forecasting. The design of xRAG4TS comprises two key components: a *Cross-City Selective Retriever Module* and an *LLM Inference Module*. Rather than transferring the entire source city, the retriever module focuses on selectively identifying and retrieving the most semantically and temporally relevant time series patterns from the source city to augment forecasting in the target city. Specifically, given a query from the target domain (e.g., a partial observation of a car park's recent availability) and abundant data from the source domain, xRAG4TS pretrains a temporal encoder based on the Masked AutoEncoder (MAE)[35] to generate representation embeddings. These embeddings are used to compute similarity scores and retrieve informative source context and sequences, which are then formatted into contextual prompts for the downstream LLM-based inference. This targeted retrieval strategy ensures that only transferable and context-relevant knowledge is incorporated, thereby improving both accuracy and robustness in low-data regimes.

The LLM Inference Module distinguishes xRAG4TS from conventional black-box transfer learning methods by enabling in-context reasoning through powerful pretrained language models. We utilize

GPT-o1¹ to perform inference over the target city inputs and retrieved source contexts, using its pretrained world knowledge and temporal reasoning capabilities to analyze correlations between future ground truth with historical records and retrieved domain-specific semantics. We then fine-tune an LLM to further learn global temporal dynamics, assisted by the learned correlations in a supervised fine-tuning (SFT) manner [42]. To operationalize this at scale, we adapt GPT-o1's reasoning ability into a smaller and more efficient LLaMA-3.1-8B model². This SFT stage adapts the LLM to the cross-city time series domain while preserving interpretability and predictive consistency. As a result, xRAG4TS achieves lightweight and deployable inference without requiring access to proprietary large-scale models, while maintaining strong generalization and explanation capabilities.

Importantly, xRAG4TS is designed to be plug-and-play compatible with existing transfer learning frameworks. By incorporating the predictions from a base transfer model (e.g., TransGTR[17]) as part of the SFT prompt context, xRAG4TS functions as an auxiliary reasoning module that complements and enhances the base model. To validate our approach, we conduct comprehensive experiments on three real-world parking availability datasets from Singapore, Nottingham, and Glasgow. Compared with state-of-the-art time series forecasting models and recent transfer learning baselines, xRAG4TS achieves substantial improvements in forecasting accuracy. We also provide detailed ablation studies and visualizations to demonstrate the impact of retrieval quality, prompt design, and model architecture. In summary, our contributions are as follows:

- We propose xRAG4TS, a novel retrieval-augmented generation framework for cross-city time series forecasting, which enables selective and interpretable knowledge transfer from source to target cities.
- We propose a novel *Cross-City Selective Retriever Module* that leverages a pretrained temporal encoder and a retrieval-augmented generation (RAG) block to dynamically select relevant and semantically similar time series, enabling effective and selective knowledge transfer.
- We design a structured *LLM Inference Module* that utilizes GPT-o1 for causal reasoning and inference on retrieved source information, the target query, and the ground truth values. We adapt the inference capabilities of GPT-o1 into a domain-adapted LLaMA-3.1-8B model through supervised fine-tuning (SFT) for enhanced forecasting performance.
- We conduct extensive experiments on real-world parking availability datasets. Experimental results show that xRAG4TS consistently outperforms competitive baselines by a large margin, validating the effectiveness, robustness, and generalizability of our approach.

2 Related Work

2.1 Time Series Transfer Learning Models

Time series forecasting has progressed from classical models like ARIMA [43] to neural approaches such as RNNs [33, 37] and Transformers [4, 53], which better capture non-linear and long-range dependencies. More recently, spatio-temporal graph neural networks

¹<https://platform.openai.com/docs/models>

²<https://huggingface.co/meta-llama/Llama-3.1-8B>

(STGNNs) [1, 20, 34, 36, 52] have advanced the field by jointly modeling spatial and temporal correlations. However, their deployment is often hindered by limited data availability in many cities, where collecting sufficient historical records requires long-term sensing infrastructure. To mitigate this, transfer learning has emerged as an effective strategy to transfer knowledge from data-rich source cities to data-scarce target cities.

Early approaches [16, 40, 50] focus on grid-based cross-city transfer learning. For instance, RegionTrans [40] employs ConvLSTM layers [37] as the backbone and utilizes auxiliary data similarity to guide the transfer process from the source to the target city. CrossTREs [16] further refines transfer granularity by decoupling node-level and edge-level knowledge and introducing a weighting network to match regions across cities during training. However, these methods still involve transferring information from all source regions, including those assigned low relevance weights, which may introduce noise and reduce transfer effectiveness. These methods also neglect the temporal dynamics and fail to retrieve the most relevant time slices and information.

Recent models [17, 27, 38] extend transfer learning to graph-structured urban data by leveraging techniques like meta-learning and transferable graph structure adaptation. For example, ST-GFSL [27] employs meta-learning to facilitate knowledge transfer across spatio-temporal graphs, while TransGTR [17] introduces a structure learning framework that learns a transferable graph constructor from the source city and applies it to the target city. Despite these advancements, such methods generally perform full-domain transfer using pre-defined or globally learned graphs, without explicitly modeling the heterogeneous relevance between source and target nodes. As a result, they may incorporate irrelevant or noisy source patterns, leading to suboptimal forecasting performance and reduced transfer effectiveness.

2.2 Large Language Models for Time Series

Recent advancements in large language models (LLMs) [2, 18, 21, 41] have expanded their applicability beyond traditional natural language processing tasks, demonstrating growing potential in time series forecasting, particularly in transfer learning and few-shot learning settings. A key strength of LLMs lies in their ability to adapt pretrained knowledge to new tasks with limited data, making them well-suited for cross-city forecasting scenarios where target domains often suffer from data scarcity and effective knowledge transfer is crucial. Several pioneering studies [3, 15, 54] have explored adapting LLMs for time series applications. For example, GPT4TS [54] and LLM4TS [3] repurpose language modeling architectures by introducing numerical tokenization and structured fine-tuning strategies, yielding moderate forecasting performance. Time-LLM [15] further advances this line of work by reprogramming temporal inputs into prompt-style formats and applying the Prompt-as-Prefix approach to enable autoregressive forecasting with LLMs. However, these methods typically overlook the spatial dependencies inherent in multivariate time series and do not incorporate knowledge retrieval and transfer to guide transfer across cities. Additionally, their reliance on purely numerical tokenization limits the model's ability to perform causal reasoning, which is essential for addressing complex, real-world forecasting scenarios.

2.3 Retrieval-Augmented Time Series Models

While large language models (LLMs) have demonstrated strong performance across various forecasting tasks, their generalization often deteriorates in the presence of non-stationary distributions, domain shifts, or limited training data, which are common conditions encountered in cross-city forecasting. Retrieval-Augmented Generation (RAG) provides a promising solution to these challenges by incorporating additional, task-relevant knowledge directly into the inference process, thereby enhancing robustness and adaptability. Recent studies [7, 22, 29, 39, 49] have begun extending RAG techniques to time series domains. For example, TimeRAG [49] retrieves time slices with high temporal similarity between the source sequence and the target sequence directly and incorporates them into a frozen LLM through a reprogramming module to guide sequence prediction. Additionally, RAF [39] and TS-RAG [29] employ parameter-frozen, pretrained encoders to embed both the query and source time series, and perform retrieval using Euclidean distances in the representation space. These models then concatenate retrieved time series with the input, then feed it to the forecasting models. Despite these advances, existing methods often rely on shallow similarity metrics or domain-agnostic embeddings from a parameter-frozen encoder, without explicitly modeling domain-specific dynamics. As a result, the retrieved sequences may include semantically or dynamically misaligned patterns, leading to suboptimal retrieval precision and reduced forecasting performance.

3 Method

Our proposed framework, xRAG4TS, is designed to enable selective, precise, and interpretable knowledge transfer for cross-city time series forecasting. The overall framework of xRAG4TS is illustrated in Fig. 1. Unlike conventional transfer learning approaches that indiscriminately transfer global models or entire graph structures, xRAG4TS focuses on identifying and leveraging only the most semantically relevant source patterns to deliver interpretable forecasting in the target city. To achieve this, xRAG4TS comprises two key components: a Cross-City Selective Retriever Module, which retrieves source information with high semantic relevance to the target query, and an LLM Inference Module, which performs structured reasoning over the retrieved source context and target sequence to generate accurate and explainable predictions.

Specifically, we construct paired samples between the source city long-term time sequences (from one week to several months, depending on the source city data spanning) and their associated contextual information. We pretrain a patch-based temporal encoder on the source city time sequence in the fashion of the Marked Autoencoder[35]. Given a target city query, the model encodes the observed sequence and retrieves top-k source sequences by computing embedding similarity against the paired source set. This process enables the model to disentangle useful domain knowledge from irrelevant signals. The corresponding contextual information and the long-term sequence information from these matched source samples are then injected into the LLM Inference Module. In the LLM Inference Module, we first employ a powerful foundation model such as GPT-o1 to perform structured reasoning and inference over the retrieved source information, the target contextual and temporal input, and the associated ground-truth label. We then

construct a supervised fine-tuning (SFT) prompt that integrates the retrieved knowledge with the target city's input, and fine-tune an LLaMA-3.1-8B model to adapt the reasoning patterns and enhance forecasting performance. This fine-tuned model operates efficiently at the inference stage while maintaining strong generalization and interpretability.

3.1 Cross-City Selective Retriever Module

In this module, we design a patch-based Masked Autoencoder (MAE) pre-training pipeline to learn domain-adaptable temporal representations from data-rich source cities, enabling selective retrieval of transferable patterns. The source city dataset typically contains long-term sequences spanning weeks or months. For each retrieval query, we slice from these long-term sequences a short-term segment whose temporal length matches the target city training span (e.g., the latest three days), ensuring a fair and contextually aligned comparison between domains. These short-term sequences are then processed as follows.

Let $\mathbf{X} \in \mathbb{R}^{B \times N \times L}$ denote a mini-batch of multivariate time series, with B samples, N nodes (e.g., car parks), and length L . Each sample is segmented into non-overlapping patches of width p , producing $L' = L/p$ patches:

$$\mathbf{P} \in \mathbb{R}^{B \times N \times L' \times p}, \quad \mathbf{P}_{b,n,t,:} = \mathbf{X}_{b,n,(t-1)p+1:t p}. \quad (1)$$

A linear projection maps each patch to d dimensions, yielding $\mathbf{E} \in \mathbb{R}^{B \times N \times L' \times d}$, followed by instance normalization to mitigate scale heterogeneity:

$$\hat{\mathbf{E}} = \frac{\mathbf{E} - \mu(\mathbf{E})}{\sigma(\mathbf{E}) + \epsilon}. \quad (2)$$

We process each node independently by folding the node axis into the batch and arranging the *patch* index and *embedding* dimension as a 2D grid. Concretely, we reshape

$$\hat{\mathbf{E}} \in \mathbb{R}^{(BN) \times 1 \times L' \times d}, \quad (3)$$

so that height $H = L'$ is the number of patches and width $W = d$ is the hidden dimension. To further enrich local temporal and feature interactions before feeding into the decoder, we apply a **spatial CNN** with padding to preserve (H, W) :

$$\mathbf{Z}' = \underbrace{\text{Conv}_{3 \times 3}^{1 \rightarrow 16} \rightarrow \text{ReLU} \rightarrow \text{Conv}_{3 \times 3}^{16 \rightarrow 16} \rightarrow \text{ReLU} \rightarrow \text{Conv}_{1 \times 1}^{16 \rightarrow 1}}_{\text{Spatial CNN}}(\hat{\mathbf{E}}). \quad (4)$$

The spatial CNN acts as a local feature refiner over the 2D patch-embedding grid to reduce noise and highlight local spatio-temporal patterns.

Finally, we reshape the output back $\tilde{\mathbf{E}} \in \mathbb{R}^{B \times N \times L' \times d}$ and feed to a stack of Transformer decoder layers. As shown in Fig. 1 upper left component, during MAE pre-training [35], a random subset $\mathcal{M} \subset \{1, \dots, L'\}$ of patches is masked; the encoder sees only visible tokens, and a lightweight decoder reconstructs masked patches back in the original time sequences. Let $\hat{\mathbf{X}}$ be the reconstruction; the loss is

$$\mathcal{L}_{\text{MAE}} = \|\mathbf{M} \odot (\hat{\mathbf{X}} - \mathbf{X})\|_1, \quad (5)$$

where $\mathbf{M}$ selects masked positions. The decoder is intentionally kept lightweight, as it is used only during MAE pre-training and

is discarded thereafter; this encourages the encoder to capture transferable temporal structure.

After pre-training, each sliced source segment is embedded by the temporal encoder to build a large-scale domain knowledge base

$$\mathcal{D} = \{(\text{id}, \mathbf{V}, \mathbf{S}, T) \mid \text{id} \in \mathcal{I}\}, \quad (6)$$

where id denotes the identifier of a specific source segment and $\mathcal{I}$ is the full set of segment identifiers in the knowledge base. In our setting, $\mathcal{D}$ contains more than 500k segment samples collected from the source city. Each entry stores $\mathbf{V} \in \mathbb{R}^{L' \times d}$, the representation embedding of segment id , $\mathbf{S}$, the corresponding long-term sequence spanning from one week to multiple weeks, and T , the associated contextual information.

Given a target query of sequence length L , the encoder produces the query embedding $\mathbf{Q} \in \mathbb{R}^{L' \times d}$. To improve retrieval precision, we compute similarity as the average of two metrics: the negative L2 distance and the negative Mahalanobis distance, both computed after flattening the matrices to vectors in $\mathbb{R}^{L'd}$. While L2 assumes isotropic covariance, Mahalanobis accounts for cross-feature correlations and is therefore more appropriate for measuring similarity in heterogeneous multivariate settings. Let $\mathbf{q} = \text{vec}(\mathbf{Q}) \in \mathbb{R}^{L'd}$ and $\mathbf{v} = \text{vec}(\mathbf{V}) \in \mathbb{R}^{L'd}$. The L2 distance is:

$$D_{\text{L2}}(\mathbf{q}, \mathbf{v}) = \sqrt{\sum_{i=1}^{L'd} (q_i - v_i)^2}, \quad (7)$$

and the Mahalanobis distance is:

$$D_{\text{Mah}}(\mathbf{q}, \mathbf{v}) = \sqrt{(\mathbf{q} - \mathbf{v})^\top \Sigma^{-1} (\mathbf{q} - \mathbf{v})}, \quad (8)$$

where Σ is the covariance matrix estimated once from all flattened embeddings in $\mathcal{D}$.

The final similarity score is, and α is a hyperparameter:

$$\text{sim}(\mathbf{Q}, \mathbf{V}) = -[\alpha D_{\text{L2}}(\mathbf{q}, \mathbf{v}) + (1 - \alpha) D_{\text{Mah}}(\mathbf{q}, \mathbf{v})]. \quad (9)$$

We then retrieve the top- K most relevant segments:

$$\mathcal{N}_K(\mathbf{Q}) = \arg \text{topK}_{(\text{id}, \mathbf{V}, \mathbf{S}, T) \in \mathcal{D}} \text{sim}(\mathbf{Q}, \mathbf{V}), \quad (10)$$

and use their $(\mathbf{S}, T)$ pairs as structured auxiliary context for the downstream LLM inference.

3.2 LLM Inference Module

The *LLM Inference Module* integrates the predictions of a spatio-temporal (ST) base forecasting model with retrieved source-domain context to perform causal reasoning over both textual and numerical modalities. This design enables context-aware correction of base-model outputs while preserving interpretability.

Pre-training of the Base ST Model. We first train a spatio-temporal base forecasting model using the training datasets from the source and target cities. The base model can be instantiated as either a classical time-series forecasting baseline or a transfer-learning backbone. In our implementation, we adopt **TransGTR** [17] due to its strong cross-city forecasting performance. Once trained, the model generates numerical forecasts for *all* training and test samples in the target city. The resulting *prediction tokens* are organized as

$$\mathbf{C}_{\text{train}} \in \mathbb{R}^{N_{\text{train}} \times N_{\text{nodes}} \times L_{\text{pred}}}, \quad \mathbf{C}_{\text{test}} \in \mathbb{R}^{N_{\text{test}} \times N_{\text{nodes}} \times L_{\text{pred}}},$$

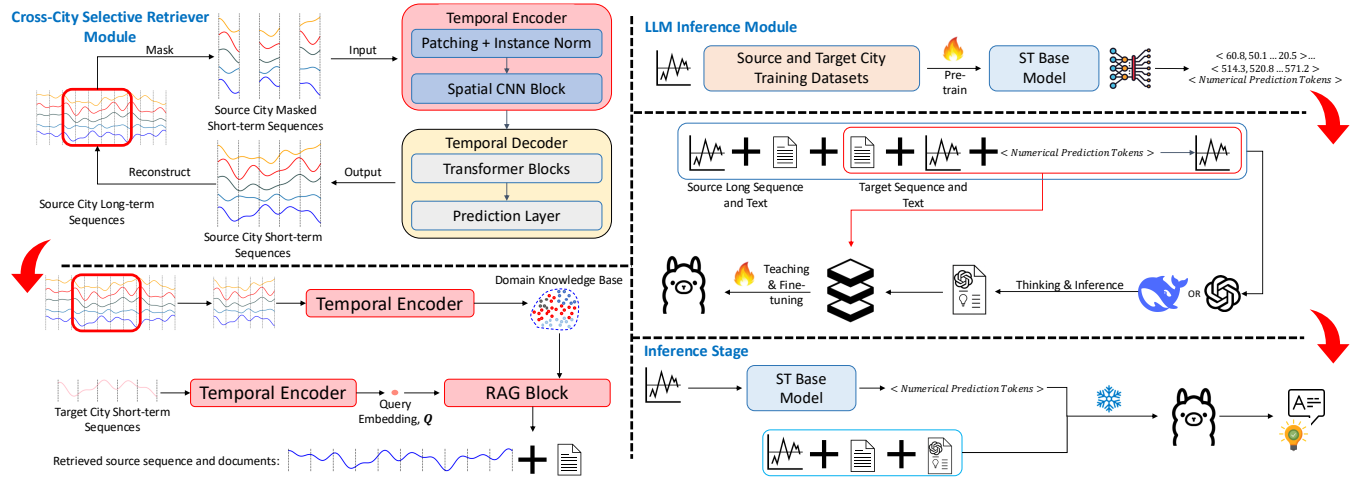


Figure 1: The overall architecture of the proposed xRAG4TS framework. The Cross-City Selective Retriever Module is on the left, and the LLM Inference Module is on the right.

where $N_{\text{train}}/N_{\text{test}}$ are the number of training/testing sequences, N_{nodes} is the number of target nodes (e.g., car parks), and L_{pred} is the forecasting horizon. Each token thus encodes the full L_{pred} -step prediction for a specific node in a specific sample, forming a structured numerical representation of the base model's output.

Correlation Extraction. To capture high-quality reasoning patterns while minimizing token consumption, we introduce a low-cost correlation extraction stage. For each target-city car park, top-K training instances are sampled. For each selected instance, we construct a structured prompt that concatenates: (1) source-domain context (textual description and the retrieved long-term source sequence); (2) target-domain context (textual description and the observed short-term target sequence); (3) the corresponding prediction tokens from C_{train} for the node relevant to that instance; and (4) the ground-truth target sequence for the forecast horizon. This prompt is fed into a reasoning-capable LLM such as GPT-o1, which performs causal reasoning to infer correlations between textual semantics, numerical forecasts, and observed ground truth. The model returns structured *hints* that distill these correlations. A prompt example is shown for better illustration.

xRAG4TS Reasoning Prompt (LLM Inference Module)

[INST] Role: You are an AI agent specialized in *cross-city transfer learning analysis* for parking availability. **Objective:** Do *not* output future parking values. Instead, use reasoning and causal inference to extract **useful, generalizable hints** that relate the inputs (textual context, source long-term sequence, target short-term history, and simulation predictions) to the **ground truth** sequence. These hints will be used later to fine-tune a student model (e.g., LLaMA-3.1-8B). Focus on: (i) bias/scale calibration of simulation predictions, (ii) lag/seasonality/day-of-week effects, (iii) cross-city pattern alignment.

Input Data:

- (1) **Source city textual information:** *Jem Shopping Mall, Singapore* — suburban mall in Jurong East, directly connected to Jurong East MRT interchange and adjacent to Westgate. 241 shops across six

levels (~818,000 sq ft). Major retailers include IKEA, FairPrice Xtra, H&M, Uniqlo, Muji, Courts, Don Don Donki. Carpark: 669 bays (B2 & B3), EV charging at B3. Pricing (incl. 9% GST from 2024-01-01): Mon–Thu first hour \$2.18, subsequent 15 mins \$0.55; Fri–Sun first hour \$2.73, subsequent 15 mins \$0.65; after 6 pm \$3.27 per entry.

- (2) **Source city long-term sequence:** Retrieved multi-week availability series with high semantic similarity to the target carpark: `<source_seq_values>`.
- (3) **Prediction horizon:** Given 3 hours of history 2017-01-24 14:30 to 2017-01-24 17:30 (Tuesday), analyze for the period 2017-01-24 17:45 to 2017-01-24 20:30.
- (4) **Target city textual information:** *St Marks Place NCP Car Park, Newark* — A spacious and affordable multi-story facility with 512 spaces, located on Lombard Street (NG24 1XT) in Newark and Sherwood. Height restriction: 1.98 m; includes dedicated disabled bays. Open 24/7 with flexible, budget-friendly pricing: £1.95 (1 h), £3.90 (2 h), £6.95 (24 h); early-bird discount before 9:00 am. Payment via ticket, card, phone, or NCP App (Location ID 32089). Known for safety (Parking Safety Award), though navigating inside can be tight due to high concrete sides. Typical patterns: peak availability 8 pm–6 am, lowest availability 10 am–3 pm.
- (5) **Target city historical records (last 3h):** `<target_seq_values>`.
- (6) **Simulation predictions (base ST model, e.g., TransGTR):** `<Prediction_tokens>`.

Training-time supervision: The ground truth for the horizon is provided as `<ground truth>`. Use it *only* to infer relationships (e.g., bias/scale, residual directions, regime shifts). **Do not** echo the ground truth or produce numeric forecasts.

Analysis Goals (what to extract):

- **Cross-city alignment:** Which parts of the source long-term sequence best align with the target history (e.g., day-of-week, hour-of-day, shopping-peak proximity, transit adjacency)? Provide a short rationale.
- **Lag/seasonality cues:** Dominant lags (e.g., 1, 2, 4 steps), diurnal phase, weekday/weekend effects, and expected monotonicity segments over the horizon.
- **Change-points/regimes:** Any shift boundaries (e.g., pre-/post-evening peak) and how they affect corrections.

Rules:

- (R1) Do *not* output any numeric forecasts or restate <ground truth>.
- (R2) Reason causally from text + sequences; prefer explanations tied to retail/transit factors and diurnal/weekday structure.

[/INST]

xRAG4TS: GPT-o1 Response

The two carparks share **similar urban contexts** — both located near high-footfall commercial areas and public transit hubs, leading to **comparable weekday commuter-driven demand peaks and evening recoveries**. By analyzing the **long-term weekly patterns from the Singapore source city**, we can infer **weekend usage profiles absent in the Newark short history**, revealing that **weekend midday periods are moderately busy but less congested than weekdays**. After **normalizing and fitting both the source and target sequences**, we **extract useful patterns and rescale to the target city scale**.

Based on the given parking lot records and a detailed observation of the available parking spaces over the week, here is a more specific breakdown of when peaks (maximum availability) and dips (minimum availability) occur: *On Weekdays (Monday to Friday):* **Peak availability: From 8 PM to 6 AM** (reaching the maximum capacity of 512 spaces). This reflects the time when parking demand is lowest, typically during the late-night and early morning hours. **Minimum availability: 10 AM to 3 PM** (223–374 spaces), indicating busiest hours likely due to work or daily activities. *On Weekends (Saturday and Sunday):* **Peak availability: From 8 PM to 6 AM** (full capacity), indicating lower demand during these hours. **Minimum availability: 11 AM to 4 PM** (248–374 spaces), showing moderate to high usage during midday hours. In summary, the parking lot **consistently reaches full capacity during late evening and early morning hours**, with the **highest usage during mid-morning to mid-afternoon** across both weekdays and weekends.

Supervised Fine-tuning of a Lightweight LLM. The final goal of the reasoning extraction stage is to distill the high-quality causal inferences from a proprietary reasoning LLM (e.g., GPT-o1) into a smaller, deployable model such as **LLaMA-3.1-8B**. This is achieved via supervised fine-tuning (SFT).

Each *SFT instance* corresponds to a single target-city forecasting case and is constructed as follows:

- (1) **Context Instruction:** The prompt contains (i) the target city’s textual description; (ii) the short-term observed target sequence; and (iii) the base ST model’s numerical prediction tokens from C_{train} containing L_{pred} forecast steps.
- (2) **Reasoning Supervision:** The GPT-o1 output that encodes patterns such as regime-shift boundaries, seasonal alignment cues, and lag dependencies.

- (3) **Ground-truth Alignment:** The ground truth values are expressed in the textual domain. This ensures the student model learns *reasoning strategies* and refines the numerical prediction tokens accordingly.

By training on this dataset, the lightweight LLaMA-3.1-8B learns to replicate GPT-o1’s reasoning behavior when given new inputs at test time. This SFT process has several advantages: (i) it significantly reduces inference cost compared to running GPT-o1 online; (ii) it considers global temporal dynamics in the target city through batched fine-tuning instead of only focusing on single temporal query, thereby, providing better forecasting performance; and (iii) it preserves interpretability, as the fine-tuned model can be prompted to generate human-readable reasoning traces.

Inference Procedure. At inference, the ground-truth future sequence is masked. The pre-trained ST base model processes the observed input sequence from the target-city test set to generate the prediction token set C_{test} , which contains forecasts for all target nodes and all L_{pred} horizon steps. During prompting, each SFT instance is constructed with only the prediction token corresponding to a single node–horizon pair, together with the target city’s context and historical sequence, and the generated structured hints. The fine-tuned LLaMA-3.1-8B then performs structured reasoning over this instance to yield the final forecast.

This two-stage design (i) integrates numerical regression (ST base model) with high-level reasoning (LLM), improving modularity and interpretability; (ii) adapts GPT-o1 style reasoning into a smaller model for efficient deployment; and (iii) unifies numerical predictions, temporal sequences, and semantic descriptions in a single inference pipeline, enabling context-aware correction of base-model errors.

4 Experiment

In this section, we evaluate the effectiveness of the proposed xRAG4TS framework for cross-city time series forecasting. We begin by detailing the experimental setup, including dataset descriptions, evaluation protocols, and baseline comparisons. To validate the advantages of xRAG4TS, we conduct a comprehensive performance comparison against state-of-the-art forecasting and transfer learning methods. We further design ablation experiments to analyze the contribution of each component in xRAG4TS, such as the retriever component and the LLM reasoning enhancement component. Additionally, we demonstrate that xRAG4TS can serve as a plug-and-play enhancement to existing forecasting backbones by selectively incorporating useful source knowledge, improving transferability and robustness in data-scarce target domains.

4.1 Experimental Setup

Datasets. We conduct experiments on three real-world parking availability datasets to evaluate the cross-city forecasting performance of our proposed xRAG4TS framework. Specifically, we use the Singapore Carpark Dataset as the source domain and two datasets, Nottingham and Glasgow, as the target domains. The Singapore Carpark Dataset contains parking availability records from 139 carparks across Singapore, recorded at 15-minute intervals. This large-scale dataset serves as the source domain due to its data abundance and high spatial coverage. The Nottingham Dataset

Table 1: Performance comparison on Nottingham dataset trained on Singapore dataset at selected horizons. Best results in red, second best in blue.

	15 mins			75 mins			135 mins			180 mins		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
ARIMA	11.16	29.37	7.06%	33.31	70.63	25.55%	57.58	107.80	50.01%	75.82	133.07	63.97%
DLinear	10.58	29.47	5.37%	32.38	68.90	22.27%	54.72	98.37	43.53%	72.70	118.43	60.12%
PatchTST	16.39	43.70	14.59%	37.80	79.67	33.03%	59.17	109.30	54.85%	75.48	130.20	73.00%
GPT4TS	9.17	30.31	6.99%	26.68	67.25	21.81%	46.79	94.71	42.82%	61.69	114.37	59.03%
Time-LLM	17.59	29.34	16.81%	39.27	78.49	37.91%	62.61	98.56	53.97%	66.16	117.74	46.12%
GTS	9.04	29.91	6.76%	24.39	62.49	19.33%	45.08	87.17	32.77%	56.37	112.23	53.37%
GWN	10.21	30.34	6.40%	26.30	61.51	22.97%	40.89	82.02	41.47%	51.10	95.30	54.79%
ST-GFSL	9.47	30.12	6.93%	24.19	62.43	18.65%	38.29	81.43	35.91%	48.37	95.13	46.27%
TPB	8.68	29.01	6.30%	23.97	62.20	18.44%	38.04	81.14	35.65%	48.08	94.82	45.98%
TransGTR	8.14	28.47	6.56%	22.77	60.81	18.50%	37.20	79.79	35.71%	46.61	93.28	44.73%
xRAG4TS	7.09	32.67	4.73%	20.44	59.30	16.36%	34.39	78.73	26.84%	44.08	93.63	35.16%

includes 15-minute interval records from 11 car parks in Nottingham, UK. The Glasgow Dataset consists of data from 7 car parks in Glasgow, UK, also recorded at 15-minute intervals. All datasets are split into training, validation, and testing sets using a standard 70%/10%/20% ratio based on chronological order. To emulate a transfer learning scenario, we restrict the training data of the target cities (Nottingham and Glasgow) to only the most recent three days available within their respective training splits, following a similar setup to [27]. We follow GTS [34] and use fixed input and output windows of $L_{in} = L_{out} = 12$ steps.

Baselines. We benchmark ten representative forecasting methods spanning statistical models, spatio-temporal graph neural networks (STGNNs), general time-series forecasters, and cross-city transfer approaches: *ARIMA*[43], *Graph WaveNet*[46], *GTS*[34], *DLinear*[25], *PatchTST*[28], *GPT4TS*[54], *TimeLLM*[15], *ST-GFSL*[27], *TPB*[26] and *TransGTR*[17]. We evaluate model performance using three standard metrics commonly employed in multivariate time series forecasting: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). Lower values across these metrics indicate better predictive accuracy.

4.2 Performance of xRAG4TS

We evaluate the performance of xRAG4TS against a diverse set of baselines and experimental results are shown in Table 1 and Table 2. Across all evaluations, xRAG4TS achieves the best performance on **10 out of 12** evaluations for Nottingham and **11 out of 12** for Glasgow, yielding a total of **21 out of 24** wins. This consistent dominance over strong baselines such as TransGTR, GTS, and GPT4TS underscores the robustness of our retrieval-augmented reasoning approach.

Nottingham results. On the Nottingham dataset (Table 1), xRAG4TS consistently outperforms all baselines across most horizons. At a 15-minute horizon, it achieves an MAE of 7.09 and MAPE of 4.73%, representing a 12.9% reduction in MAE and a 27.9% reduction in MAPE compared to TransGTR. Even at the longest horizon (180 minutes), xRAG4TS maintains clear superiority with an MAE of 44.08 and MAPE of 35.16%, outperforming TransGTR by 5.4%

and 21.4%, respectively. These results highlight its ability to mitigate long-term error propagation—a key limitation of conventional transfer learning models.

Glasgow results. A similar trend is observed for the Glasgow dataset (Table 2). At the shortest horizon (15 minutes), xRAG4TS records an MAE of 7.50 and MAPE of 1.50%, improving over TransGTR by 23.8% and 23.5%, respectively. At 180 minutes, it achieves an MAE of 45.60 and MAPE of 9.15%, outperforming TransGTR by 10.5% and 9.4%. Gains are especially pronounced at intermediate horizons (75 and 135 minutes), where xRAG4TS demonstrates greater resilience to temporal degradation compared to all baselines.

Three consistent patterns emerge across both datasets: (1) **Consistent superiority over baselines:** xRAG4TS dominates the majority of evaluations, with particularly strong margins at longer horizons. (2) **Improved prediction stability:** Compared with baselines that suffer from unstable percentage errors, xRAG4TS yields substantially lower MAPE, indicating more reliable forecasting across horizons. (3) **Synergy between retrieval and reasoning:** The performance gap between xRAG4TS and pure ST models (e.g., TransGTR, GTS) shows that integrating semantically relevant retrieved sequences with causal reasoning boosts predictive accuracy beyond what parameter transfer alone can achieve. By combining selective retrieval of domain-relevant sequences with LLM-driven inference, xRAG4TS effectively leverages transferable knowledge while avoiding negative transfer effects. This enables superior accuracy across both short- and long-term horizons, establishing xRAG4TS as a robust and generalizable solution for cross-city time series forecasting. To provide an intuitive understanding of the forecasting behavior and the reasoning capability of our proposed framework, we provide a detailed real-world case study and visualization in the Appendix B. We also evaluate the efficiency of our proposed framework against two state-of-the-art reasoning models, GPT-o1 and DeepSeekR1, and report in the Appendix C.

4.3 Ablation Study

To assess the contribution of each core component in **xRAG4TS**, we design two controlled variants of the model on the Nottingham dataset, each modifying a single module while keeping all other

Table 2: Performance comparison on Glasgow dataset trained on Singapore dataset at selected horizons. Best results in red, second best in blue.

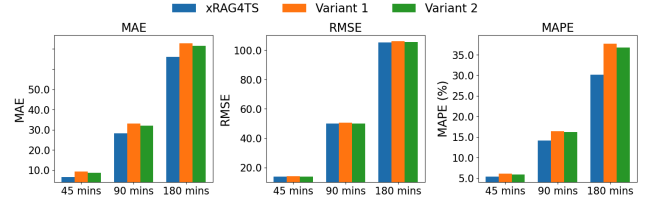
	15 mins			75 mins			135 mins			180 mins		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
ARIMA	12.88	32.33	2.49%	41.37	78.79	7.41%	73.74	131.85	12.88%	99.62	173.92	16.66%
DLinear	10.62	34.06	1.89%	32.92	83.82	6.58%	57.13	122.56	11.73%	81.83	147.10	16.18%
PatchTST	20.52	55.05	3.95%	44.69	99.38	8.91%	68.86	136.35	13.92%	86.77	160.69	17.85%
GPT4TS	9.93	34.82	1.98%	29.06	81.29	5.83%	51.91	117.11	10.71%	69.35	141.49	14.50%
TimeLLM	10.32	35.78	2.12%	31.24	85.93	6.63%	55.21	120.24	12.54%	75.75	148.21	16.50%
GTS	10.28	34.38	2.76%	26.34	69.60	5.24%	44.61	92.96	8.41%	58.35	114.80	10.21%
GWN	12.93	36.30	2.35%	29.74	68.01	5.30%	42.38	92.24	8.22%	52.09	105.49	10.14%
ST-GFSL	10.48	33.01	2.11%	26.34	68.80	5.22%	41.05	95.02	8.12%	51.73	108.91	10.28%
TPB	10.25	32.62	2.01%	26.07	68.41	5.08%	40.81	94.70	7.98%	51.41	108.45	10.03%
TransGTR	9.84	32.05	1.96%	25.53	67.60	4.89%	39.51	91.27	7.86%	50.92	105.23	10.10%
xRAG4TS	7.50	31.95	1.50%	23.41	66.04	4.64%	37.47	90.86	7.51%	45.60	105.99	9.15%

settings identical to the original. The performance of these variants is compared to the full **xRAG4TS** system, and the results are visualized in Fig. 2.

Variant 1: Random-Centroid Retrieval. In this variant, we study the role of the retriever module by replacing the similarity-based retrieval mechanism with a k -means clustering over the source-city embeddings. For each cluster, a single centroid sequence and its associated textual description are randomly selected as the long-term context for all target queries assigned to that cluster. This process removes fine-grained semantic matching between source and target sequences, potentially degrading the quality of retrieved patterns.

Variant 2: Open-Source LLM Inference. In this variant, we evaluate the importance of the reasoning capacity in the LLM Inference Module by replacing the proprietary LLMs (GPT-o1, DeepSeek-R1) with a directly deployed open-source **LLaMA-3.1-8B** model. The same input format is preserved, but without the distilled high-quality reasoning traces from GPT-o1, the reasoning process is expected to be less nuanced.

Analysis. Both ablation settings result in a noticeable drop in forecasting accuracy, particularly in MAE and MAPE, while RMSE changes are relatively smaller. This indicates that these components are especially effective for improving forecasting performance, whereas RMSE is less affected. For Variant 1, the performance drop confirms that semantic retrieval from the source domain is critical for providing representative and domain-relevant long-term patterns, which directly enhance the downstream LLM reasoning stage. To further validate the effectiveness of the proposed selective retriever, we present two representative case studies with detailed visualizations in Appendix B. The results show that the retriever consistently identifies semantically relevant and temporally aligned patterns from the source city, thereby ensuring that only the most informative knowledge is transferred to enhance forecasting performance in the target city. For Variant 2, the weaker performance shows that while LLaMA-3.1-8B can perform some correction, it lacks the advanced cross-city causal inference capabilities of GPT-o1, validating the effectiveness of our two-stage reasoning–distillation approach.

**Figure 2: Ablation study results on the Nottingham dataset.**

5 Conclusion

In this work, we present xRAG4TS, a retrieval-augmented framework for cross-city time series forecasting that selectively transfers relevant knowledge from the data-rich source cities to improve forecasting accuracy in data-scarce target cities. By combining a Cross-City Selective Retriever Module, xRAG4TS filters out irrelevant knowledge and enhances transfer effectiveness. Additionally, we incorporate an LLM Inference Module, where GPT-o1’s reasoning capability is adapted into a compact LLaMA-3.1-8B model via supervised fine-tuning. Our method is plug-and-play compatible with existing transfer frameworks, such as TransGTR, and significantly boosts their performance. Extensive experiments on real-world parking availability datasets across multiple cities validate the effectiveness and generalizability of xRAG4TS. Results show that our framework consistently outperforms existing baselines, highlighting the potential of retrieval-based transfer and LLM-driven forecasting in spatio-temporal domains. Future work includes extending xRAG4TS to multi-modal urban data sources from multi-domains and investigating scalable retrieval mechanisms for larger temporal corpora.

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A Implementation Details

In this section, we introduce the details of our implementation. All experiments are conducted in PyTorch on a Linux workstation equipped with an Intel® Core™ i7-13700K CPU and a single NVIDIA A6000 GPU (48 GB). We enable FP16 mixed-precision training and gradient checkpointing to reduce memory usage. The LLaMA-3.1-8B model is quantized to 4-bit [23] and adapted using LoRA [48] with a rank of $r=64$. We use an initial learning rate of 2×10^{-4} with the Adam optimizer. xRAG4TS is fine-tuned for 10 epochs using LLaMA-3.1-8B, while TransGTR [17] is pre-trained as the base model following its official default configuration. The mask ratio used in the xRAG4TS retriever module is 0.75, the hyperparameter, α , is 0.75, and the long sequence length from the source city is set to 14 days with the details listed in Table 3. In our xRAG4TS experiments, we set top-K to be 1. All baselines and our xRAG4TS model were evaluated on the aforementioned GPU, except for ARIMA, which was fitted solely on the test-domain dataset using the CPU.

For evaluation, we adopt three standard metrics widely used in multivariate time series forecasting: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). Lower values across all metrics indicate better predictive accuracy. The metrics are defined as:

Mean Absolute Error (MAE).

$$MAE = \frac{1}{N_{\text{sample}}} \sum_{i=1}^{N_{\text{sample}}} |y_i - \hat{y}_i|, \quad (11)$$

Root Mean Squared Error (RMSE).

$$RMSE = \sqrt{\frac{1}{N_{\text{sample}}} \sum_{i=1}^{N_{\text{sample}}} (y_i - \hat{y}_i)^2}, \quad (12)$$

Mean Absolute Percentage Error (MAPE).

$$MAPE = \frac{100}{N_{\text{sample}}} \sum_{i=1}^{N_{\text{sample}}} \left| \frac{y_i - \hat{y}_i}{y_i} \right|, \quad (13)$$

where y_i denotes the ground-truth value at time step i , $\hat{y}_i$ is the corresponding predicted value, and N_{sample} is the total number of forecasted samples.

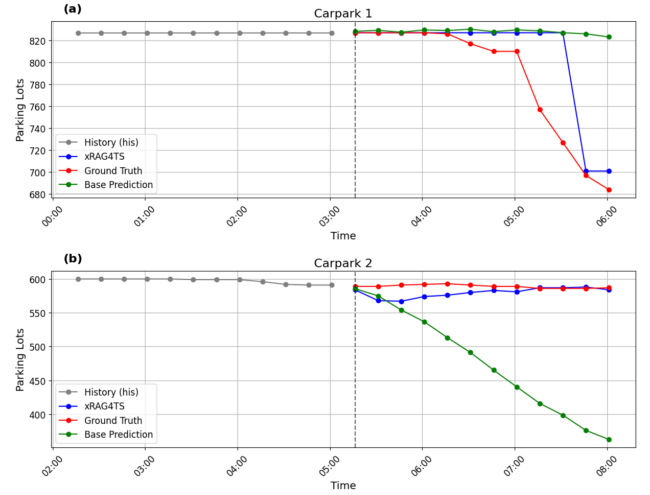


Figure 3: Visualization of forecasting results. (a) For Carpark 1, xRAG4TS accurately captures the decline in parking lot availability and follows the ground truth closely. (b) Compared with the base model TransGTR, xRAG4TS correctly identifies that the parking availability remains stable, avoiding the erroneous sharp decrease predicted by TransGTR. These examples illustrate the effectiveness and robustness of our proposed framework.

B Case Study and Visualization

To provide an intuitive understanding of the forecasting behavior of our proposed framework, we present visualization results in Fig. 3. As shown in Fig. 3(a), for Carpark 1, xRAG4TS is able to recognize the decline in parking lot availability and closely follow the ground-truth trend, demonstrating its ability to capture dynamic variations. In contrast, Fig. 3(b) highlights a scenario where the baseline model TransGTR incorrectly predicts a sharp decrease in availability, while xRAG4TS successfully identifies that the parking availability remains relatively stable. These visualizations clearly indicate that xRAG4TS not only adapts to diverse temporal dynamics but also avoids overestimating drastic fluctuations, thereby producing more reliable and accurate forecasts.

Furthermore, we highlight that xRAG4TS in the inference stage is capable of addressing the black-box limitation by distilling the high-quality causal inferences from a proprietary reasoning LLM.

Table 3: Statistics of the real-world datasets.

Datasets	Time Frequency	Time range	Total Parking Records
Singapore	15 mins	1 May 2021 - 6 June 2021	813,984
Nottingham	15 mins	26 Oct 2016 - 16 Feb 2017	119,108
Glasgow	15 mins	26 Oct 2016 - 16 Feb 2017	119,108

Table 4: Comparison of xRAG4TS with GPT-o1 and DeepSeekR1 in terms of model parameters, inference GPU usage, and token-level costs.

Model	Model Parameter	Inference GPU Usage	Input Cost (\$/M tokens)	Output Cost (\$/M tokens)
xRAG4TS	8B	~11 GB	0.3 to 0.45	0.3 to 0.45
GPT-o1	unknown	unknown	15	60
DeepSeekR1	671B (37B MoE)	1500 GB	0.55	2.19

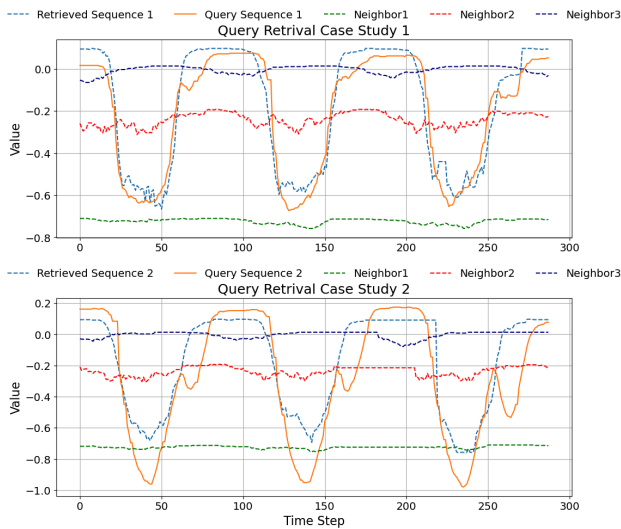


Figure 4: Case studies demonstrating the effectiveness of the retriever module. In both the top and bottom examples, the retrieved source sequence (after z-normalization) matches the target query sequence closely, while neighboring sequences deviate more noticeably. This confirms that the retriever accurately identifies semantically relevant patterns to support robust cross-city forecasting.

Beyond faithfully aligning with the simulation trajectory and historical context, xRAG4TS enables causal reasoning and structured inference that allow for seamless extension of the forecasting horizon without any parameter updates. For instance, in our prompt-based case study corresponding to Fig. 3(a), xRAG4TS not only refined its 3-hour forecasts by discounting simulation fluctuations and incorporating domain-specific commuting patterns, but also demonstrated the ability to continue forecasting for the subsequent hour. Such forward-looking reasoning, grounded in retrieved knowledge and structured hints, cannot be achieved by the baseline method, underscoring the unique interpretability and adaptability of our retrieval-augmented framework. Furthermore, the alignment between the ground truth values (dropping to 600 to

700 lots) and the reasoning provided by xRAG4TS further validates its interpretability. Overall, these case studies confirm the effectiveness of our retrieval-augmented design in improving forecasting robustness and fidelity. Regarding our data integration strategy, we utilize both textual metadata and time-series signals within a unified framework. While space constraints precluded the reporting of single-modality experiments, prior works utilizing only text [41] or derived temporal descriptors [15] have demonstrated the utility of unimodal inputs. However, these approaches are typically designed for self-performance enhancement rather than cross-city generalization and lack the capacity for interpretable, step-by-step reasoning.

To further validate the effectiveness of our selective retriever, we illustrate two representative case studies in Fig. 4. In each case, the top and bottom panels correspond to different query sequences from the target domain. After applying z-normalization, the retrieved source sequence shows a strong alignment with the query sequence, whereas the neighboring sequences deviate more significantly. This demonstrates that our retriever is capable of capturing semantically relevant and temporally consistent patterns from the source city, ensuring that the most informative knowledge is transferred to support forecasting in the target city.

xRAG4TS Inference Stage

```
[INST] Instruction: [INST] Role: You are an AI agent
[details omitted]...
```

Consider all provided information (historical records, simulation predictions, and additional factors) to predict parking lot availability for the next 3 hours with accuracy. [Formatting details omitted].[/INST]

```
xRAG4TS.: [827. 827. 827. 827. 827. 827. 827. 827. 827.
827. 701. 701.]
```

Users: Can you provide some reasons why you adjusted your answer from the referenced tokens?

xRAG4TS: The past 3 hours of observed records were a perfectly flat sequence at **827**, which matches the car park’s known late-night “reset state” (nearly full availability before the morning commute). The LLM Inference Module thus weighted these stable ground truths more heavily than noisy simulation fluctuations.

From the domain knowledge, availability consistently drops between **07:00–09:00**. The input query horizon (06:46–09:31) lies exactly in this interval. Therefore, instead of continuing the flat 827 values, the inference adjusted sharply downward to **701** for the final two intervals, reflecting the start of morning arrivals. This drop is consistent with retrieved Nottingham patterns showing 100–150 car arrivals in this slot on weekdays.

Users: I changed my mind can you also provide next hour values for me, from 2017-02-09 09:46:14 to 2017-02-09 10:31:14?

xRAG4TS: Based on xRAG4TS’s reasoning mechanisms and the retrieved cross-city priors, the **next hour (2017-02-09 09:46:14 → 10:31:14)** forecasts would continue the morning drop trend but at a moderated rate, as arrivals slow slightly after the main 07:00–09:00 surge: Predicted availability for the next 4 intervals (15-min each): **[669. 657. 626. 621.]**

C Efficiency Comparison with GPT-o1 and DeepSeekR1

To further evaluate the efficiency of our proposed framework, we compare xRAG4TS with two state-of-the-art reasoning models, GPT-o1 and DeepSeekR1. As reported in Table 4, when executed on an NVIDIA A100 80GB GPU during inference, xRAG4TS exhibits a pronounced efficiency advantage. We estimate these inference costs based on the A100 inference speed reported in [32], combined with pricing from commonly used online GPU renting platforms such as Lambda and AutoDL.

While GPT-o1 and DeepSeekR1 rely on extremely large-scale models (hundreds of billions of parameters) and require massive GPU resources—notably up to 1500 GB VRAM for DeepSeekR1 according to technical hardware reports—xRAG4TS operates with a lightweight 8B backbone that requires only about 11 GB GPU memory for inference when loaded with LoRA. This efficiency enables xRAG4TS to run cost-free in our self-hosted setup, whereas GPT-o1 and DeepSeekR1 incur substantial API costs, with GPT-o1 charging up to \$60 per million output tokens and DeepSeekR1 charging \$2.19 per million output tokens. The GPT-o1 API calls used to generate reasoning traces are considered a one-time offline cost (totaling under \$20) and are therefore excluded from the inference-time analysis. All inference components in xRAG4TS employ open-source models, and the SFT stage incurs no additional API expenses.

Moreover, xRAG4TS is able to capture diverse temporal dynamics through parameter updates, thereby delivering robust performance that cannot be achieved by merely invoking GPT-o1 or DeepSeekR1 API [11]. Furthermore, we report the per-sample inference latency for the cross-city transfer task (SG → Nottingham). The process comprises three stages: the TransGTR backbone inference (0.35s), the retrieval phase (0.37s), and the xRAG4TS generation phase (2.51s with a 1200-token limit). Consequently, the total latency for xRAG4TS is 3.23s, which compares favorably to the GPT-o1 baseline inference time of 3.91s. In practical terms, this means that xRAG4TS achieves robust reasoning and forecasting capabilities at a fraction of the resource footprint, making it substantially more suitable for large-scale or real-time deployments where inference efficiency is critical.

D Performance on the ECL Dataset

To evaluate the effectiveness of xRAG4TS, we adopt the experimental setup from FSTLLM [12]. Specifically, we focus on forecasting the averaged hourly electricity consumption (measured in kilowatt-hours) for 19 target clients within the Electricity Consumption Load (ECL) dataset. Following the established protocol, the remaining 300 clients serve as the source dataset to facilitate cross-client knowledge transfer. The quantitative results are summarized in Table 5. Comparison against the strong baseline, TransGTR, reveals that xRAG4TS achieves superior forecasting performance across all evaluated metrics. Most notably, our model demonstrates a substantial improvement in Mean Absolute Percentage Error (MAPE), reducing it by 24.8%. This significant drop indicates that xRAG4TS is highly robust to scale variations and captures the relative consumption dynamics more accurately than the baseline. Additionally, we observe a solid 8.07% reduction in MAE, further confirming the model’s precision in tracking ground-truth electricity usage. These results validate that retrieving relevant temporal patterns from the source clients effectively mitigates the distribution shift in the target domain, leading to more reliable energy demand forecasts.

Table 5: Performance on the ECL dataset (measured in kilowatt-hours).

	MAE	MAPE	RMSE
TransGTR	2792.83	23.8%	5150.37
xRAG4TS	2567.33	17.9%	5087.11
Improvements	8.07%	24.8%	1.23%